MODULE 3: DISTRIBUTED DATABASES

UNIT 5: DATA WAREHOUSING

3.5.0 Introduction

Large companies have presences in many places, each of which may generate a large volume of data. For instance, large retail chains have hundreds or thousands of stores, whereas insurance companies may have data from thousands of local branches. Further, large organizations have a complex internal organization structure, and therefore different data may be present in different locations, or on different operational systems, or under different schemas. For instance, manufacturing-problem data and customer-complaint data may be stored on different database systems. Organizations often purchase data from external sources, such as mailing lists that are used for product promotions, or credit scores of customers that are provided by credit bureaus, to decide on credit-worthiness of customers. Corporate decision makers require access to information from multiple such sources. Setting up queries on individual sources is both cumbersome and inefficient. Moreover, the sources of data may store only current data, whereas decision makers may need access to past data as well; for instance, information about how purchase patterns have changed in the past year could be of great importance. Data warehouses provide a solution to these problems.

3.5.1 Objectives

At the end of this unit, students should be able to:

1. To understand data warehousing as an active decision support framework
2. To understand the architecture and components of data ware houses
3. To understand data ware housing schema

3.5.1.1 What is Data Warehouse?

A data warehouse is a repository (or archive) of information gathered from multiple sources, stored under a unified schema, at a single site. A data warehouse is simply a single, complete, and consistent store of data obtained from a variety of sources and made available to end users in a way they can understand and use it in a business context. Once gathered, the data are stored for a long time, permitting access to historical data. Thus, data warehouses provide the user a single consolidated interface to data, making decision-support queries easier to write. Moreover, by accessing information for decision support from a data warehouse, the decision maker ensures that online transaction-processing systems are not affected by the decision-support workload. *A data warehouse is a subject-oriented, integrated, time variant, and nonvolatile collection of data in support of management's decision-making process.*

Subject Oriented:

1. Organized around major subjects, such as customer, product, sales.
2. Focusing on the modeling and analysis of data for decision makers, not on daily operations or transaction processing.
3. Provide a simple and concise view around particular subject issues by excluding data that are not useful in the decision support process.

Integrated

1. Data on a given subject is defined and stored once
2. Constructed by integrating multiple, heterogeneous data sources – relational databases, flat files, on-line transaction records
3. One set of consistent, accurate, quality information
4. Standardization
   - Naming conventions
   - Coding structures
   - Data attributes
   - Measures

1. Data cleaning and data integration techniques are applied.
   - Ensure consistency in naming conventions, encoding structures, attribute measures, etc. among different data sources
   - When data is moved to the warehouse, it is converted

Time Variant

Data is stored as a series of snapshots, each representing a period of time

The time horizon for the data warehouse is significantly longer than that of operational systems.

- – Operational database: current value data
- – Data warehouse data: provide information from a historical perspective (example, past 5-10 years)
- • **Every key structure in the data warehouse**
- – Contains an element of time, explicitly or implicitly
- – But the key of operational data may or may not contain "time element"

Non-Volatile

1. Typically data in the data warehouse is not deleted
2. A physically separate store of data transformed from the operational environment
3. Operational update of data does not occur in the data warehouse environment
4. Does not require transaction processing, recovery, and concurrency control mechanisms
5. Requires only two operations in data accessibility

3.5.2 Components of a Data Warehouse

Figure 3.5.1 shows the architecture of a typical data warehouse, and illustrates the gathering of data, the storage of data, and the querying and data analysis support. Among the issues to be addressed in building a warehouse are the following:

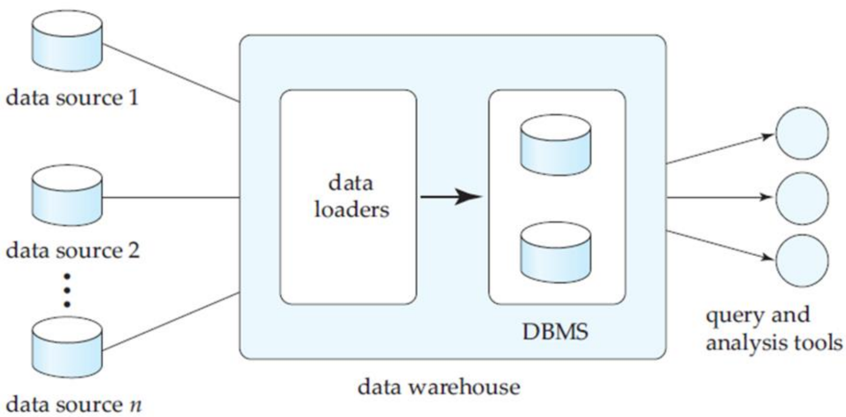

Figure 3.5.1: Architecture of a Data Warehouse

1. **When and how to gather data.** In a **source-driven architecture** for gathering data, the data sources transmit new information, either continually (as transaction processing takes place), or periodically (nightly, for example). In a **destination-driven architecture**, the data warehouse periodically sends requests for new data to the sources. Unless updates at the sources are replicated at the warehouse via two phase commit, the warehouse will never be quite up-to-date with the sources. Two-phase commit is usually far too expensive to be an option, so data warehouses typically have slightly *out-of-date* data. That, however, is usually not a problem for decision-support systems.
2. **What schema to use.** Data sources that have been constructed independently are likely to have different schemas. In fact, they may even use different data models. Part of the task of a warehouse is to perform schema integration, and to convert data to the integrated schema before they are stored. As a result, the data stored in the warehouse are not just a copy of the data at the sources. Instead, they can be thought of as a materialized view of the data at the sources.

3. **Data transformation and cleansing.** The task of correcting and preprocessing data is called **data cleansing**. Data sources often deliver data with numerous minor inconsistencies, which can be corrected. For example, names are often misspelled, and addresses may have street, area, or city names misspelled, or postal codes entered incorrectly. These can be corrected to a reasonable extent by consulting a database of street names and postal codes in each city. The approximate matching of data required for this task is referred to as **fuzzy lookup**.

4. Address lists collected from multiple sources may have duplicates that need to be eliminated in a **merge–purge operation** (this operation is also referred to as

deduplication). Records for multiple individuals in a house may be grouped together so only one mailing is sent to each house; this operation is called ***householding***.

5. Data may be **transformed** in ways other than cleansing, such as changing the units of measure, or converting the data to a different schema by joining data from multiple source relations. Data warehouses typically have graphical tools to support data transformation. Such tools allow transformation to be specified as boxes, and edges can be created between boxes to indicate the flow of data. Conditional boxes can route data to an appropriate next step in transformation.

6. **How to propagate updates**. Updates on relations at the data sources must be propagated to the data warehouse. If the relations at the data warehouse are exactly the same as those at the data source, the propagation is straightforward.

7. **What data to summarize.** The raw data generated by a transaction-processing system may be too large to store online. However, we can answer many queries by maintaining just summary data obtained by aggregation on a relation, rather than maintaining the entire relation. For example, instead of storing data about every sale of clothing, we can store total sales of clothing by item name and category.

8. The different steps involved in getting data into a data warehouse are called **extract, transform, and load** or **ETL** tasks; extraction refers to getting data from the sources, while load refers to loading the data into the data warehouse.

3.5.3 Warehouse Schemas

Data warehouses typically have schemas that are designed for data analysis, using tools such as OLAP tools. Thus, the data are usually multidimensional data, with dimension attributes and measure attributes. Tables containing multidimensional data are called **fact tables** and are usually very large. A table recording sales information for a retail store, with one tuple for each item that is sold, is a typical example of a fact table. The dimensions of the *sales* table would include what the item is (usually an item identifier such as that used in bar codes), the date when the item is sold, which location (store) the item was sold from, which customer bought the item, and so on. The measure attributes may include the number of items sold and the price of the items.

To minimize storage requirements, dimension attributes are usually short identifiers that are foreign keys into other tables called **dimension tables**. For instance, a fact table *sales* would have attributes *item id*, *store id*, *customer id*, and *date*, and measure attributes *number* and *price*. The attribute *store id* is a foreign key into a dimension table *store*, which has other attributes such as store location (city, state, country). The *item id* attribute of the *sales* table would be a foreign key into a dimension table *item info*, which would contain information such as the name of the item, the category to which the item belongs, and other item details such as color and size. The *customer id* attribute would be a foreign key into a *customer* table containing attributes such as name and address of the customer. We can also view the *date* attribute as a foreign key into a *date info* table giving the month, quarter, and year of each date.

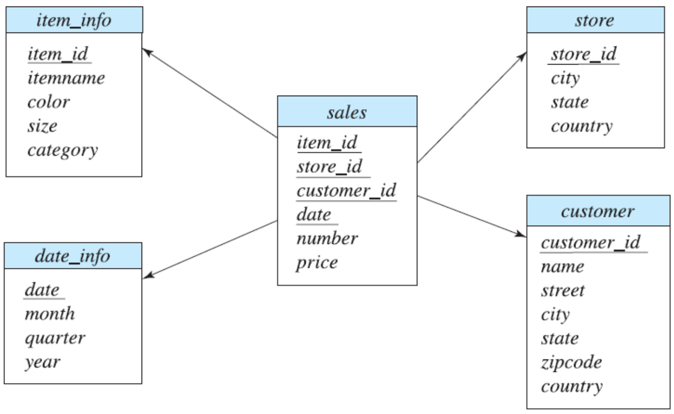

Figure 3.5.2: Star schema fora Data Warehouse

The resultant schema appears in Figure 3.5.2. Such a schema, with a fact table, multiple dimension tables, and foreign keys from the fact table to the dimension tables, is called a **star schema**. More complex data-warehouse designs may have multiple levels of dimension tables; for instance, the *item info* table may have an attribute *manufacturer id* that is a foreign key into another table giving details of the manufacturer. Such schemas are called **snowflake schemas**. Complex data warehouse designs may also have more than one fact table.

3.5.4 Conclusion:

Decision-support systems analyze online data collected by transaction-processing systems, to help people make business decisions. Since most organizations are extensively computerized today, a very large body of information is available for decision support. Decision-support systems come in various forms, including OLAP systems and data-mining systems. Data warehouses help gather and archive important operational data. Warehouses are used for decision support and analysis on historical data, for instance, to predict trends. Data cleansing from input data sources is often a major task in data warehousing. Warehouse schemas tend to be multidimensional, involving one or a few very large fact tables and several much smaller dimension tables

3.5.5 Tutor Marked Assignment

1. What is data Warehouse?
2. Describe the main characteristics of a star schema
3. With suitable illustration describe the components of a data warehouse

3.5.6 References and Further Readings

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